Class 20: Maxwell-Boltzmann Distribution Vs Fermi-Dirac Distribution

In the previous class we looked at the plot of the Fermi-Dirac distribution function, and the features of the function.

In this class we will compare the Maxwell-Boltzmann distribution with the Fermi-Dirac distribution.

The Maxwell-Boltzmann distribution and its variation with temperature are shown in Figure 20.1 below.

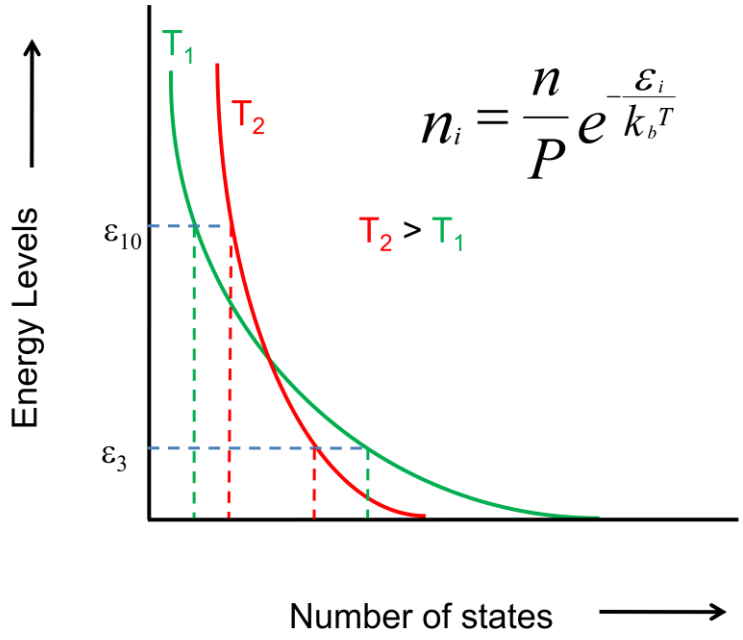


Figure 20.1: Maxwell-Boltzmann distribution function, and its variation with temperature

At higher temperatures, higher energy levels get increased numbers of electrons, while the lower energy levels get decreased numbers of electrons.

The Maxwell-Boltzmann distribution function can be represented in a normalized manner as follows:

$$P(\varepsilon) = P_0 e^{\frac{-\varepsilon}{k_b T}}$$

Such that

$$\int_0^\infty P(\varepsilon) d\,\varepsilon = 1$$

This normalized function is typically plotted with its axes interchanged with respect to the axes in Figure 20.1, i.e. with energy on the x-axis and $P(\varepsilon)$ on the y-axis.

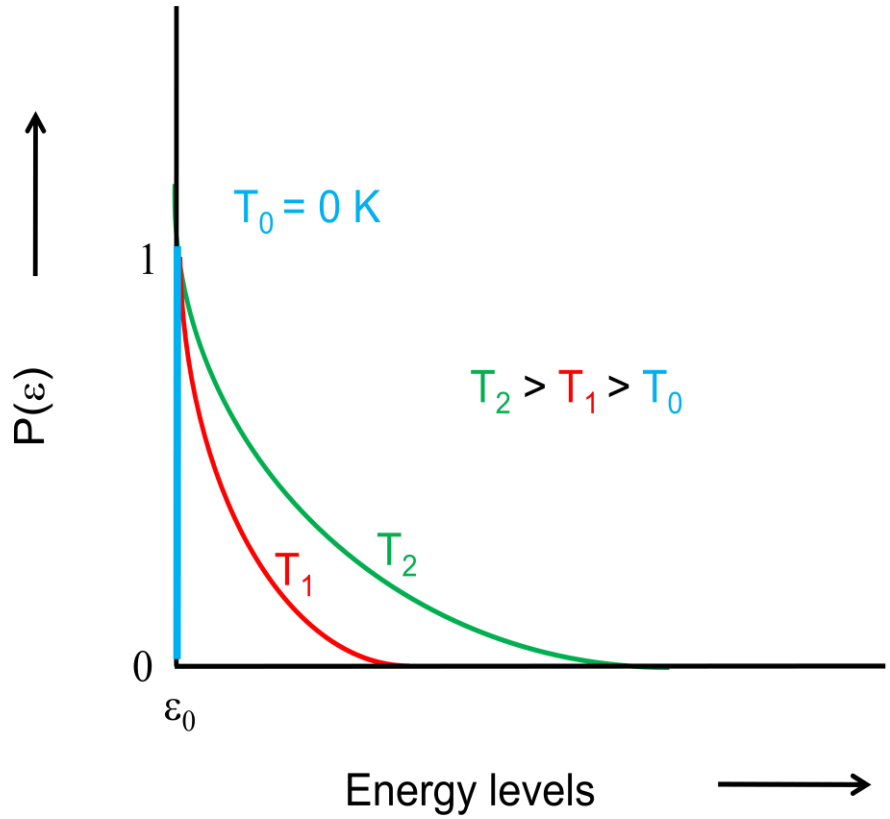


Figure 20.2: Plot of the normalized Maxwell-Boltzmann distribution function, as a function of energy at different temperatures

The normalized Maxwell-Boltzmann distribution function coincides with the y-axis at T=0 Kelvin. At higher temperatures, higher energy levels are occupied as indicated in the curves in Figure 20.2 above.

The Maxwell-Boltzmann distribution overestimates C_V^e. The reason for this is evident from the plot of the normalized distribution shown above. As the temperature increases, all of the electrons can access empty states at energy levels higher than the energy level that they occupy. Therefore all of them can, in principle, participate in the process of increasing the temperature, by gaining energy. Since there are a large number of electrons in the system, the energy required to raise the temperature is large, and this is directly reflected in the value of the specific heat, which is a measure of this energy.

The Fermi-Dirac distribution creates a situation wherein at lower energy levels all of the states are full. Therefore, when an electron at a relatively low energy level tries to gain energy and move to a marginally higher energy level, it is unable to do so since all of the states at that marginally higher energy level are already full. This situation is indicated illustratively in Figure 20.3 below.

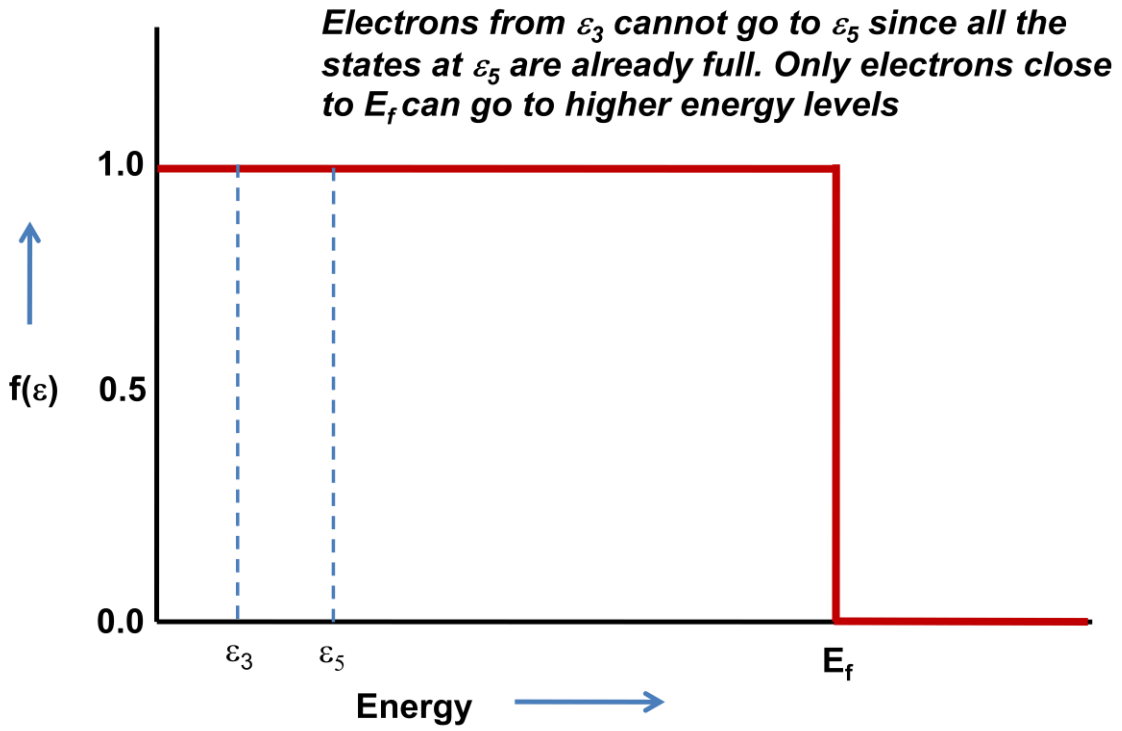


Figure 20.3: An illustrative example showing how lower energy levels are unable to participate in the process of gaining energy consistent with an increased temperature of the system. Electrons at energy level ε_3 are unable to gain energy and go to states at energy level ε_5, since those states are already full. Only electrons close to Fermi energy are in a position to gain energy and go to unoccupied states at higher energy levels.

Only electrons close to the Fermi energy level are able to access empty states at energy levels marginally higher than those they already occupy. Therefore only electrons close to the Fermi energy, participate in the gain in energy as the temperature increases. Since this is a very small fraction of the electrons in the system, the energy required to raise the temperature is relatively small, and this is directly reflected in the smaller value of the specific heat, which is a measure of this energy. The limitation in the number of electrons that can participate in the process of raising the energy of the system, limits the C_V^e. This situation is analogous to water evaporating only from the surface of a container that is full of water.

We are therefore able to see how, by applying the Fermi-Dirac distribution we are able to correct one of the major anomalies of the Classical Drude model.

Figure 20.4 below compares the important features of the Maxwell-Boltzmann distribution with those of the Fermi-Dirac distribution.

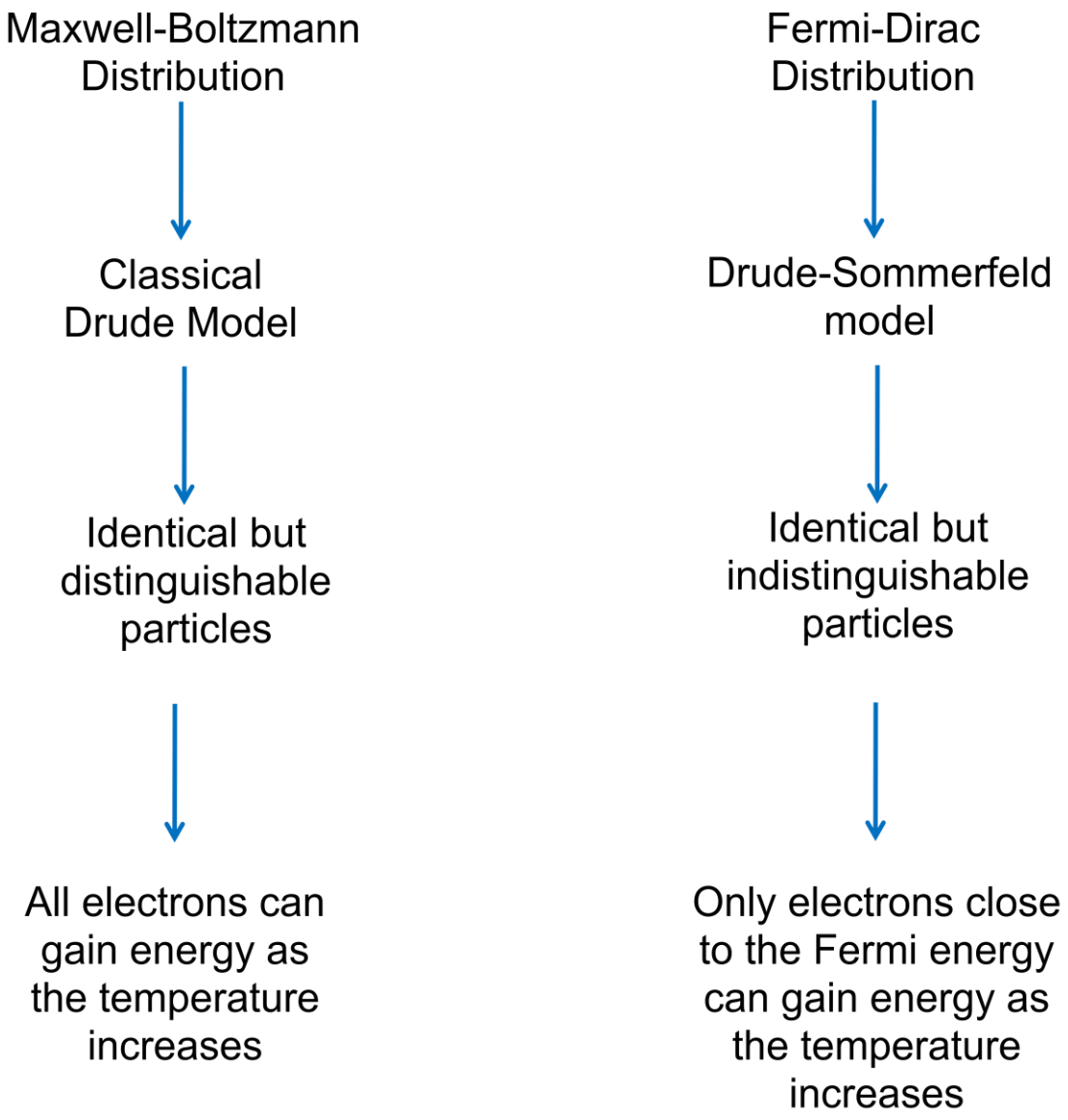


Figure 20.4: A comparison of some of the important features of the Maxwell-Boltzmann distribution with those of the Fermi-Dirac distribution

To further refine the model we have to take a fresh look at the assertion that the potential within the solid is flat, or uniform. In reality the potential that an electron will experience close to an ionic core will be significantly different from the potential it will experience far from the ionic core. We therefore need to build a picture of the potential as a function of position in a solid. It will turn out that we will be able to approximate the potential as a function of position in a solid in a manner that makes it convenient for subsequent computations. We will also look to obtain an expression for the number of states at a given energy level in the upcoming classes.